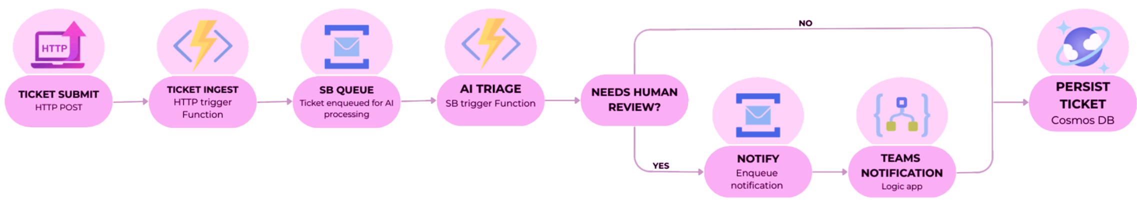
PROJECT OVERVIEW

Ticket Triage Ai is a cloud-based application that automates the classification and prioritization of support tickets using AI.
The system processes incoming tickets through an event-driven pipeline, where each request is analyzed and enriched with structured metadata such as category, severity and confidence score.

Key Features

- AI-powered ticket classification (category, severity, confidence score)
- Event-driven processing pipeline using Azure Service Bus
- Asynchronous and scalable architecture
- Persistent storage with Azure Cosmos DB
- Web dashboard for ticket submission and visualization
- Monitoring and logging with Application Insights



DATA FLOW

→ User submits a ticket from the dashboard
→ The request is validated by the Ingest API
→ A processor function consumes the message
→ The ticket is classified using Azure OpenAI
→ The result is stored in Cosmos DB
→ If the confidence score is low or the ticket requires attention, a message is sent to a review queue
→ A Logic App processes the review request and sends a notification to Microsoft Teams

DESIGN CHOICES

Event-driven architecture
Decouples ingestion, processing and notification workflows, improving scalability and flexibility

Serverless functions
Enable automatic scaling and reduce infrastructure management

Service Bus

Provides reliable messaging, buffering and supports multiple processing stages (including human review)

AI integration
Automates classification while allowing fallback to human review when confidence is low

Logic App for notifications
Separates business logic from communication workflows and enables easy integration with external systems like Microsoft Teams

IMPLEMENTATIONS NOTES

Dependency Injection
Services are registered through dependency injection to keep the application modular testable and easy to extend

Factory pattern
Factories are used to create normalized messages and persistence models, keeping transformation logic separated from the processing flow

Validation layer
Incoming requests are validated before entering the pipeline to ensure data consistency and reduce invalid processing

Global exception middleware
A middleware handles unhandled exceptions centrally, improving error management and observability

Correlation ID
Correlation IDs are propagated across the pipeline to support end-to-end tracing and debugging

Human-in-the-loop review
Low-confidence classifications are routed to review workflow through Service Bus, Logic App and Teams notifications

Dead-letter handling
Failed messages can be moved to a dead-letter flow for reliability and failure tracking